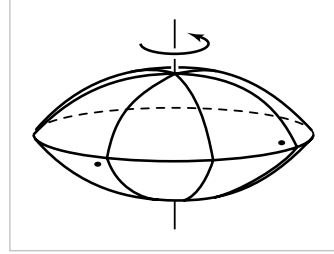


To obtain the CW structure, first subdivide the unit circle  $C$  in the  $n^{\text{th}}$   $\mathbb{C}$  factor of  $\mathbb{C}^n$  by taking the points  $e^{2\pi i j/m} \in C$  as vertices,  $j = 1, \dots, m$ . Joining the  $j^{\text{th}}$  vertex of  $C$  to the unit sphere  $S^{2n-3} \subset \mathbb{C}^{n-1}$  by arcs of great circles in  $S^{2n-1}$  yields a  $(2n-2)$ -dimensional ball  $B_j^{2n-2}$  bounded by  $S^{2n-3}$ . Specifically,  $B_j^{2n-2}$  consists of the points  $\cos \theta (0, \dots, 0, e^{2\pi i j/m}) + \sin \theta (z_1, \dots, z_{n-1}, 0)$  for  $0 \leq \theta \leq \pi/2$ . Similarly, joining the  $j^{\text{th}}$  edge of  $C$  to  $S^{2n-3}$  gives a ball  $B_j^{2n-1}$  bounded by  $B_j^{2n-2}$  and  $B_{j+1}^{2n-2}$ , subscripts being taken mod  $m$ . The rotation  $\rho$  carries  $S^{2n-3}$  to itself and rotates  $C$  by the angle  $2\pi \ell_n/m$ , hence  $\rho$  permutes the  $B_j^{2n-2}$ 's and the  $B_j^{2n-1}$ 's. A suitable power of  $\rho$ , namely  $\rho^r$  where  $r\ell_n \equiv 1 \pmod{m}$ , takes each  $B_j^{2n-2}$  and  $B_j^{2n-1}$  to the next one. Since  $\rho^r$  has order  $m$ , it is also a generator of the rotation group  $\mathbb{Z}_m$ , and hence we may obtain  $L$  as the quotient of one  $B_j^{2n-1}$  by identifying its two faces  $B_j^{2n-2}$  and  $B_{j+1}^{2n-2}$  together via  $\rho^r$ .

In particular, when  $n = 2$ ,  $B_j^{2n-1}$  is a lens-shaped 3-ball and  $L$  is obtained from this ball by identifying its two curved disk faces via  $\rho^r$ , which may be described as the composition of the reflection across the plane containing the rim of the lens, taking one face of the lens to the other, followed by a rotation of this face through the angle  $2\pi \ell/m$  where  $\ell = r\ell_1$ . The figure illustrates the case  $(m, \ell) = (7, 2)$ , with the two dots indicating a typical pair of identified points in the upper and lower faces of the lens. Since the lens space  $L$  is determined by the rotation angle  $2\pi \ell/m$ , it is conveniently written  $L_{\ell/m}$ . Clearly only the mod  $m$  value of  $\ell$  matters. It is a classical theorem of Reidemeister from the 1930s that  $L_{\ell/m}$  is homeomorphic to  $L_{\ell'/m'}$  iff  $m' = m$  and  $\ell' \equiv \pm \ell^{\pm 1} \pmod{m}$ . For example, when  $m = 7$  there are only two distinct lens spaces  $L_{1/7}$  and  $L_{2/7}$ . The 'if' part of this theorem is easy: Reflecting the lens through a mirror shows that  $L_{\ell/m} \approx L_{-\ell/m}$ , and by interchanging the roles of the two  $\mathbb{C}$  factors of  $\mathbb{C}^2$  one obtains  $L_{\ell/m} \approx L_{\ell^{-1}/m}$ . In the converse direction,  $L_{\ell/m} \approx L_{\ell'/m'}$  clearly implies  $m = m'$  since  $\pi_1(L_{\ell/m}) \approx \mathbb{Z}_m$ . The rest of the theorem takes considerably more work, involving either special 3-dimensional techniques or more algebraic methods that generalize to classify the higher-dimensional lens spaces as well. The latter approach is explained in [Cohen 1973].



Returning to the construction of a CW structure on  $L_m(\ell_1, \dots, \ell_n)$ , observe that the  $(2n-3)$ -dimensional lens space  $L_m(\ell_1, \dots, \ell_{n-1})$  sits in  $L_m(\ell_1, \dots, \ell_n)$  as the quotient of  $S^{2n-3}$ , and  $L_m(\ell_1, \dots, \ell_n)$  is obtained from this subspace by attaching two cells, of dimensions  $2n-2$  and  $2n-1$ , coming from the interiors of  $B_j^{2n-1}$  and its two identified faces  $B_j^{2n-2}$  and  $B_{j+1}^{2n-2}$ . Inductively this gives a CW structure on  $L_m(\ell_1, \dots, \ell_n)$  with one cell  $e^k$  in each dimension  $k \leq 2n-1$ .

The boundary maps in the associated cellular chain complex are computed as follows. The first one,  $d_{2n-1}$ , is zero since the identification of the two faces of  $B_j^{2n-1}$  is via a reflection (degree  $-1$ ) across  $B_j^{2n-1}$  fixing  $S^{2n-3}$ , followed by a rota-

tion (degree +1), so  $d_{2n-1}(e^{2n-1}) = e^{2n-2} - e^{2n-2} = 0$ . The next boundary map  $d_{2n-2}$  takes  $e^{2n-2}$  to  $me^{2n-3}$  since the attaching map for  $e^{2n-2}$  is the quotient map  $S^{2n-3} \rightarrow L_m(\ell_1, \dots, \ell_{n-1})$  and the balls  $B_j^{2n-3}$  in  $S^{2n-3}$  which project down onto  $e^{2n-3}$  are permuted cyclically by the rotation  $\rho$  of degree +1. Inductively, the subsequent boundary maps  $d_k$  then alternate between 0 and multiplication by  $m$ .

Also of interest are the infinite-dimensional lens spaces  $L_m(\ell_1, \ell_2, \dots) = S^\infty / \mathbb{Z}_m$  defined in the same way as in the finite-dimensional case, starting from a sequence of integers  $\ell_1, \ell_2, \dots$  relatively prime to  $m$ . The space  $L_m(\ell_1, \ell_2, \dots)$  is the union of the increasing sequence of finite-dimensional lens spaces  $L_m(\ell_1, \dots, \ell_n)$  for  $n = 1, 2, \dots$ , each of which is a subcomplex of the next in the cell structure we have just constructed, so  $L_m(\ell_1, \ell_2, \dots)$  is also a CW complex. Its cellular chain complex consists of a  $\mathbb{Z}$  in each dimension with boundary maps alternately 0 and  $m$ , so its reduced homology consists of a  $\mathbb{Z}_m$  in each odd dimension.

In the terminology of §1.B, the infinite-dimensional lens space  $L_m(\ell_1, \ell_2, \dots)$  is an Eilenberg–MacLane space  $K(\mathbb{Z}_m, 1)$  since its universal cover  $S^\infty$  is contractible, as we showed there. By Theorem 1B.8 the homotopy type of  $L_m(\ell_1, \ell_2, \dots)$  depends only on  $m$ , and not on the  $\ell_i$ 's. This is not true in the finite-dimensional case, when two lens spaces  $L_m(\ell_1, \dots, \ell_n)$  and  $L_m(\ell'_1, \dots, \ell'_n)$  have the same homotopy type iff  $\ell_1 \cdots \ell_n \equiv \pm k^n \ell'_1 \cdots \ell'_n \pmod{m}$  for some integer  $k$ . A proof of this is outlined in Exercise 2 in §3.E and Exercise 29 in §4.2. For example, the 3-dimensional lens spaces  $L_{1/5}$  and  $L_{2/5}$  are not homotopy equivalent, though they have the same fundamental group and the same homology groups. On the other hand,  $L_{1/7}$  and  $L_{2/7}$  are homotopy equivalent but not homeomorphic.

### Euler Characteristic

For a finite CW complex  $X$ , the **Euler characteristic**  $\chi(X)$  is defined to be the alternating sum  $\sum_n (-1)^n c_n$  where  $c_n$  is the number of  $n$ -cells of  $X$ , generalizing the familiar formula *vertices* – *edges* + *faces* for 2-dimensional complexes. The following result shows that  $\chi(X)$  can be defined purely in terms of homology, and hence depends only on the homotopy type of  $X$ . In particular,  $\chi(X)$  is independent of the choice of CW structure on  $X$ .

|| **Theorem 2.44.**  $\chi(X) = \sum_n (-1)^n \text{rank } H_n(X)$ .

Here the **rank** of a finitely generated abelian group is the number of  $\mathbb{Z}$  summands when the group is expressed as a direct sum of cyclic groups. We shall need the following fact, whose proof we leave as an exercise: If  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is a short exact sequence of finitely generated abelian groups, then  $\text{rank } B = \text{rank } A + \text{rank } C$ .

**Proof of 2.44:** This is purely algebraic. Let

$$0 \rightarrow C_k \xrightarrow{d_k} C_{k-1} \rightarrow \cdots \rightarrow C_1 \xrightarrow{d_1} C_0 \rightarrow 0$$

be a chain complex of finitely generated abelian groups, with cycles  $Z_n = \text{Ker } d_n$ , boundaries  $B_n = \text{Im } d_{n+1}$ , and homology  $H_n = Z_n/B_n$ . Thus we have short exact sequences  $0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$  and  $0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0$ , hence

$$\text{rank } C_n = \text{rank } Z_n + \text{rank } B_{n-1}$$

$$\text{rank } Z_n = \text{rank } B_n + \text{rank } H_n$$

Now substitute the second equation into the first, multiply the resulting equation by  $(-1)^n$ , and sum over  $n$  to get  $\sum_n (-1)^n \text{rank } C_n = \sum_n (-1)^n \text{rank } H_n$ . Applying this with  $C_n = H_n(X^n, X^{n-1})$  then gives the theorem.  $\square$

For example, the surfaces  $M_g$  and  $N_g$  have Euler characteristics  $\chi(M_g) = 2 - 2g$  and  $\chi(N_g) = 2 - g$ . Thus all the orientable surfaces  $M_g$  are distinguished from each other by their Euler characteristics, as are the nonorientable surfaces  $N_g$ , and there are only the relations  $\chi(M_g) = \chi(N_{2g})$ .

### Split Exact Sequences

Suppose one has a retraction  $r: X \rightarrow A$ , so  $ri = \mathbb{1}$  where  $i: A \rightarrow X$  is the inclusion. The induced map  $i_*: H_n(A) \rightarrow H_n(X)$  is then injective since  $r_*i_* = \mathbb{1}$ . From this it follows that the boundary maps in the long exact sequence for  $(X, A)$  are zero, so the long exact sequence breaks up into short exact sequences

$$0 \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \rightarrow 0$$

The relation  $r_*i_* = \mathbb{1}$  actually gives more information than this, by the following piece of elementary algebra:

**Splitting Lemma.** For a short exact sequence  $0 \rightarrow A \xrightarrow{i} B \xrightarrow{j} C \rightarrow 0$  of abelian groups the following statements are equivalent:

(a) There is a homomorphism  $p: B \rightarrow A$  such that  $pi = \mathbb{1}: A \rightarrow A$ .

(b) There is a homomorphism  $s: C \rightarrow B$  such that  $js = \mathbb{1}: C \rightarrow C$ .

(c) There is an isomorphism  $B \approx A \oplus C$  making

a commutative diagram as at the right, where the maps in the lower row are the obvious ones,

$a \mapsto (a, 0)$  and  $(a, c) \mapsto c$ .

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \begin{array}{c} \nearrow i \\ \searrow \end{array} & B & \begin{array}{c} \nwarrow j \\ \searrow \end{array} & C \longrightarrow 0 \\ & & & & \downarrow \approx & & \\ & & & & A \oplus C & & \end{array}$$

If these conditions are satisfied, the exact sequence is said to **split**. Note that (c) is symmetric: There is no essential difference between the roles of  $A$  and  $C$ .

**Sketch of Proof:** For the implication (a)  $\Rightarrow$  (c) one checks that the map  $B \rightarrow A \oplus C$ ,  $b \mapsto (p(b), j(b))$ , is an isomorphism with the desired properties. For (b)  $\Rightarrow$  (c) one uses instead the map  $A \oplus C \rightarrow B$ ,  $(a, c) \mapsto i(a) + s(c)$ . The opposite implications (c)  $\Rightarrow$  (a) and (c)  $\Rightarrow$  (b) are fairly obvious. If one wants to show (b)  $\Rightarrow$  (a) directly, one can define  $p(b) = i^{-1}(b - sj(b))$ . Further details are left to the reader.  $\square$

Except for the implications (b)  $\Rightarrow$  (a) and (b)  $\Rightarrow$  (c), the proof works equally well for nonabelian groups. In the nonabelian case, (b) is definitely weaker than (a) and (c), and short exact sequences satisfying (b) only determine  $B$  as a semidirect product of  $A$  and  $C$ . The difficulty is that  $s(C)$  might not be a normal subgroup of  $B$ . In the nonabelian case one defines ‘splitting’ to mean that (b) is satisfied.

In both the abelian and nonabelian contexts, if  $C$  is free then every exact sequence  $0 \rightarrow A \xrightarrow{i} B \xrightarrow{j} C \rightarrow 0$  splits, since one can define  $s: C \rightarrow B$  by choosing a basis  $\{c_\alpha\}$  for  $C$  and letting  $s(c_\alpha)$  be any element  $b_\alpha \in B$  such that  $j(b_\alpha) = c_\alpha$ . The converse is also true: If every short exact sequence ending in  $C$  splits, then  $C$  is free. This is because for every  $C$  there is a short exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  with  $B$  free — choose generators for  $C$  and let  $B$  have a basis in one-to-one correspondence with these generators, then let  $B \rightarrow C$  send each basis element to the corresponding generator — so if this sequence  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  splits,  $C$  is isomorphic to a subgroup of a free group, hence is free.

From the Splitting Lemma and the remarks preceding it we deduce that a retraction  $r: X \rightarrow A$  gives a splitting  $H_n(X) \approx H_n(A) \oplus H_n(X, A)$ . This can be used to show the nonexistence of such a retraction in some cases, for example in the situation of the Brouwer fixed point theorem, where a retraction  $D^n \rightarrow S^{n-1}$  would give an impossible splitting  $H_{n-1}(D^n) \approx H_{n-1}(S^{n-1}) \oplus H_{n-1}(D^n, S^{n-1})$ . For a somewhat more subtle example, consider the mapping cylinder  $M_f$  of a degree  $m$  map  $f: S^n \rightarrow S^n$  with  $m > 1$ . If  $M_f$  retracted onto the  $S^n \subset M_f$  corresponding to the domain of  $f$ , we would have a split short exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_n(S^n) & \longrightarrow & H_n(M_f) & \longrightarrow & H_n(M_f, S^n) \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{m} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_m \longrightarrow 0 \end{array}$$

But this sequence does not split since  $\mathbb{Z}$  is not isomorphic to  $\mathbb{Z} \oplus \mathbb{Z}_m$  if  $m > 1$ , so the retraction cannot exist. In the simplest case of the degree 2 map  $S^1 \rightarrow S^1$ ,  $z \mapsto z^2$ , this says that the Möbius band does not retract onto its boundary circle.

### Homology of Groups

In §1.B we constructed for each group  $G$  a CW complex  $K(G, 1)$  having a contractible universal cover, and we showed that the homotopy type of such a space  $K(G, 1)$  is uniquely determined by  $G$ . The homology groups  $H_n(K(G, 1))$  therefore depend only on  $G$ , and are usually denoted simply  $H_n(G)$ . The calculations for lens spaces in Example 2.43 show that  $H_n(\mathbb{Z}_m)$  is  $\mathbb{Z}_m$  for odd  $n$  and 0 for even  $n > 0$ . Since  $S^1$  is a  $K(\mathbb{Z}, 1)$  and the torus is a  $K(\mathbb{Z} \times \mathbb{Z}, 1)$ , we also know the homology of these two groups. More generally, the homology of finitely generated abelian groups can be computed from these examples using the Künneth formula in §3.B and the fact that a product  $K(G, 1) \times K(H, 1)$  is a  $K(G \times H, 1)$ .

Here is an application of the calculation of  $H_n(\mathbb{Z}_m)$ :

**Proposition 2.45.** *If a finite-dimensional CW complex  $X$  is a  $K(G, 1)$ , then the group  $G = \pi_1(X)$  must be torsionfree.*

This applies to quite a few manifolds, for example closed surfaces other than  $S^2$  and  $\mathbb{RP}^2$ , and also many 3-dimensional manifolds such as complements of knots in  $S^3$ .

**Proof:** If  $G$  had torsion, it would have a finite cyclic subgroup  $\mathbb{Z}_m$  for some  $m > 1$ , and the covering space of  $X$  corresponding to this subgroup of  $G = \pi_1(X)$  would be a  $K(\mathbb{Z}_m, 1)$ . Since  $X$  is a finite-dimensional CW complex, the same would be true of its covering space  $K(\mathbb{Z}_m, 1)$ , and hence the homology of the  $K(\mathbb{Z}_m, 1)$  would be nonzero in only finitely many dimensions. But this contradicts the fact that  $H_n(\mathbb{Z}_m)$  is nonzero for infinitely many values of  $n$ .  $\square$

Reflecting the richness of group theory, the homology of groups has been studied quite extensively. A good starting place for those wishing to learn more is the textbook [Brown 1982]. At a more advanced level the books [Adem & Milgram 1994] and [Benson 1992] treat the subject from a mostly topological viewpoint.

## Mayer-Vietoris Sequences

In addition to the long exact sequence of homology groups for a pair  $(X, A)$ , there is another sort of long exact sequence, known as a **Mayer-Vietoris sequence**, which is equally powerful but is sometimes more convenient to use. For a pair of subspaces  $A, B \subset X$  such that  $X$  is the union of the interiors of  $A$  and  $B$ , this exact sequence has the form

$$\begin{aligned} \cdots \rightarrow H_n(A \cap B) \xrightarrow{\Phi} H_n(A) \oplus H_n(B) \xrightarrow{\Psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \rightarrow \cdots \\ \cdots \rightarrow H_0(X) \rightarrow 0 \end{aligned}$$

In addition to its usefulness for calculations, the Mayer-Vietoris sequence is also applied frequently in induction arguments, where one might know that a certain statement is true for  $A$ ,  $B$ , and  $A \cap B$  by induction and then deduce that it is true for  $A \cup B$  by the exact sequence.

The Mayer-Vietoris sequence is easy to derive from the machinery of §2.1. Let  $C_n(A + B)$  be the subgroup of  $C_n(X)$  consisting of chains that are sums of chains in  $A$  and chains in  $B$ . The usual boundary map  $\partial: C_n(X) \rightarrow C_{n-1}(X)$  takes  $C_n(A + B)$  to  $C_{n-1}(A + B)$ , so the  $C_n(A + B)$ 's form a chain complex. According to Proposition 2.21, the inclusions  $C_n(A + B) \hookrightarrow C_n(X)$  induce isomorphisms on homology groups. The Mayer-Vietoris sequence is then the long exact sequence of homology groups associated to the short exact sequence of chain complexes formed by the short exact sequences

$$0 \rightarrow C_n(A \cap B) \xrightarrow{\varphi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A + B) \rightarrow 0$$